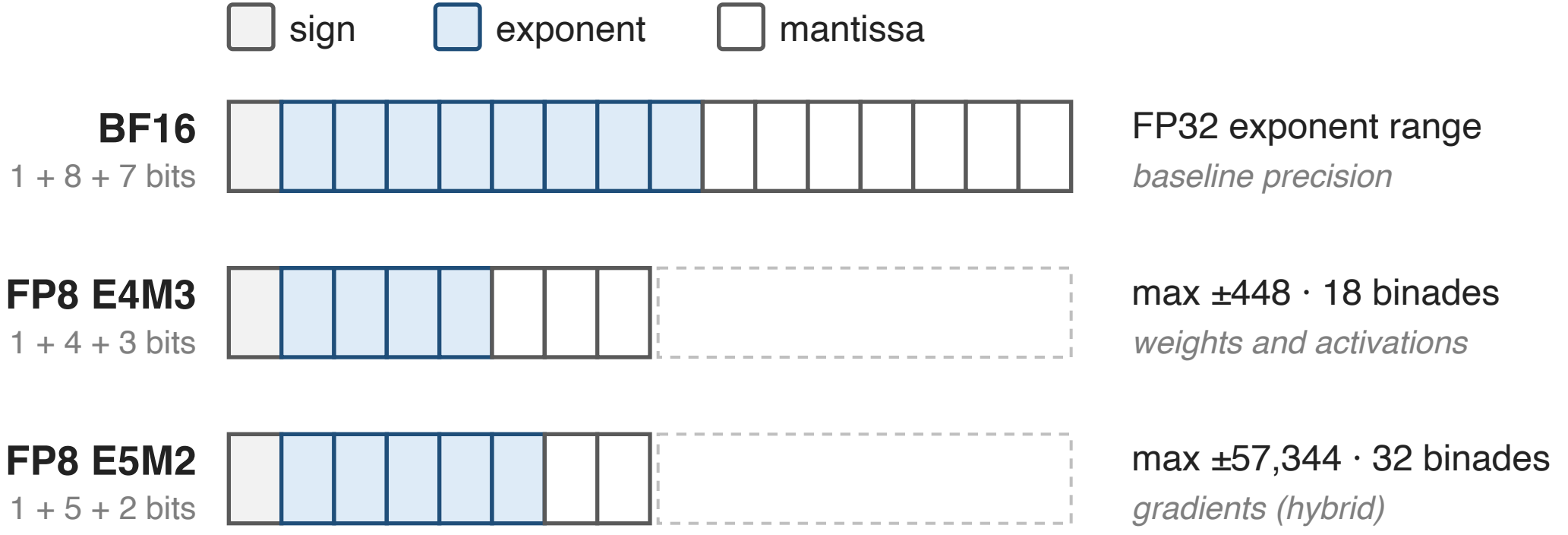
sign
exponent
mantissa
BF16
1 + 8 + 7 bits
FP32 exponent range
baseline precision
FP8 E4M3
1 + 4 + 3 bits
max ±448 · 18 binades
weights and activations
FP8 E5M2
1 + 5 + 2 bits
max ±57,344 · 32 binades
gradients (hybrid)